

GeoExplorer: Active Geo-localization with Curiosity-Driven Exploration

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<https://limirs.github.io/GeoExplorer/>

Abstract

Active Geo-localization (AGL) is the task of localizing a goal, represented in various modalities (e.g., aerial images, ground-level images, or text), within a predefined search area. Current methods approach AGL as a goal-reaching reinforcement learning (RL) problem with a distance-based reward. They localize the goal by implicitly learning to minimize the relative distance from it. However, when distance estimation becomes challenging or when encountering unseen targets and environments, the agent exhibits reduced robustness and generalization ability due to the less reliable exploration strategy learned during training. In this paper, we propose GeoExplorer, an AGL agent that incorporates curiosity-driven exploration through intrinsic rewards. Unlike distance-based rewards, our curiosity-driven reward is goal-agnostic, enabling robust, diverse, and contextually relevant exploration based on effective environment modeling. These capabilities have been proven through extensive experiments across four AGL benchmarks, demonstrating the effectiveness and generalization ability of GeoExplorer in diverse settings, particularly in localizing unfamiliar targets and environments.

1. Introduction

Guiding unmanned aerial vehicles (UAVs) is essential in search-and-rescue operations, which necessitate UAVs to screen affected areas and locate missing individuals or damaged structures [16, 19]. Instead of conducting exhaustive searches across entire regions, UAVs can be actively guided using available information: one could use coordinates of the point of interest, historical aerial imagery, or even ground photographs or textual descriptions [21, 25].

The problem is known in computer vision as Active Geo-Localization (AGL) [21, 25] within a goal-reaching reinforcement learning (RL) context (Figure 1 (a)). AGL focuses on localizing a target (*goal*), within a predefined search area (*environment*) presented in the bird’s eye view, by navigating the agent towards it. At a given time, the agent observes a *state*, i.e., a patch representing a limited

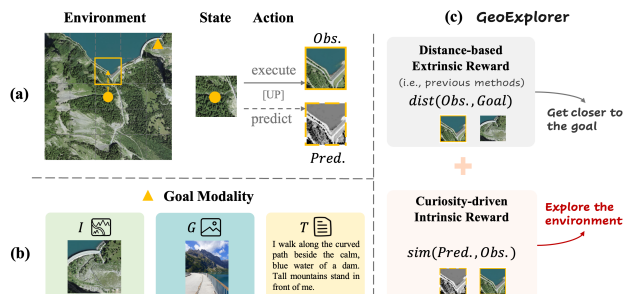


Figure 1. **GeoExplorer combines goal-oriented and curiosity-driven rewards to address AGL.** (a) *Task setting.* Given a starting patch (○) and a predefined search area (*environment*), AGL aims at localizing a goal (△), by guiding an agent towards it. At a given time, the agent observes a *state* and selects an *action*. The goal location is *unknown* for inference. (b) *Goal modalities.* The goal can be presented in various modalities including aerial images (I), ground-level images (G) and text (T), associated with a patch in the area. (c) *GeoExplorer.* The learning process of GeoExplorer is guided by two rewards: **1)** a goal-oriented **distance-based** reward, as in previous methods, which directs the agent toward the goal, and **2)** a goal-agnostic, curiosity-driven reward that encourages the agent to explore the environment based on *curiosity* measured by **similarity** of state prediction and observation.

observation of the environment, and selects an *action*, i.e., a decision that modifies the agent position and the observed state. The goal content can be described in various modalities (e.g., aerial patches, ground-level images, and textual descriptions, Figure 1 (b)), but its location is *unknown* during inference. The task is considered successful if the agent reaches the goal location within a limited search budget ($\mathcal{B}$). In real-world applications of AGL, robustness and generalizability are crucial as few scenarios are identical [25].

Existing AGL methods rely on extrinsic rewards (i.e., goal-oriented), and the policy is implicitly shaped by minimizing the relative distance between the agent and the goal during training. However, this distance is *unknown* during inference (since the goal’s location is unknown), making the learned strategy less reliable. Moreover, these distance estimations are often tailored to specific training environments, making it challenging for models to adapt to unseen

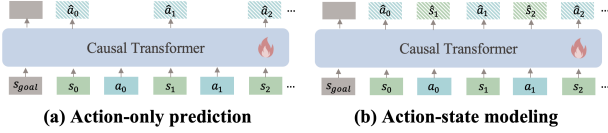


Figure 2. Comparison between (a) action-only prediction and (b) joint action-state modeling.

goals or new environments. Curiosity-driven intrinsic rewards (*i.e.*, goal-agnostic) offer a compelling solution. By rewarding the agent for encountering less familiar [6, 20] or unpredictable [20, 30] state transitions, the intrinsic reward encourages *exploration of the environment* [14, 20]. Therefore, integrating curiosity-driven rewards with goal-oriented rewards introduces an essential trade-off between following the direct, goal-oriented guidance and engaging in exploratory behavior. This balance ensures that the agent not only efficiently reaches its goals but also continually refines its understanding of the environment, leading to effective and adaptive policies which enhance *robustness and generalization* in AGL.

However, constructing intrinsic rewards requires a way to *predict* state transitions, as it is based on the discrepancy between state prediction and actual observation. Previous AGL methods [21, 25] only focus on predicting the action sequence that leads to the goal [7] (Figure 2 (a)), and therefore do not satisfy this requirement. Without explicit state prediction, these methods also fail to anticipate how an action can change the state accurately, resulting in reduced adaptability and accumulated errors in unseen environments. Instead, joint action-state modeling (Figure 2 (b)) encourages a more robust and adaptive environment modeling and representation refinement [3, 32], which is crucial when facing unfamiliar environments or goals.

In this work, we introduce GeoExplorer, an AGL agent that 1) jointly predicts the *action-state dynamics* and 2) explores the search region with a *curiosity-driven reward*. The learning process of GeoExplorer boils down to two stages: *Action-State Dynamics Modeling (DM)* and *Curiosity-Driven Exploration (CE)*. DM simultaneously models action-state dynamics using a causal Transformer and allows GeoExplorer not only to learn which actions lead to the goal, but also how these actions influence state transitions, thus providing a richer, more comprehensive representation of the environment. Then, during CE, an intrinsic reward is introduced to guide the exploration process by leveraging the discrepancy between the state prediction provided by DM and the real-world state observation. Experimental results on four datasets, including a *novel unseen target generalization* setting, demonstrate that GeoExplorer provides dense, goal-agnostic and content-aware rewards for AGL. Combined with goal-oriented guidance, GeoExplorer highlights that balancing *goal-reaching* (extrinsic re-

ward) and *environment exploration* (intrinsic reward) can improve both localization performance and generalization ability of the AGL agent. The resulting active and adaptive exploration ability of GeoExplorer represents a step towards deploying AGL in real-world search-and-rescue operations.

2. Related Work

Active Geo-Localization. Although related, AGL differs from traditional visual geo-localization [39] (*i.e.*, typically focuses on identifying the location of a target through coordinate prediction [2, 31] or retrieval [4, 5, 17, 33, 35, 37, 41]), by actively navigating the agent to reach the target. Current methods in AGL have made impressive progress in learning the action transitions from various UAV localization trajectories [21, 25]. Pirinen *et al.* [10] framed the AGL as a goal-reaching RL problem, guided by a sparse extrinsic reward. Sarkar *et al.* [25] extended the AGL task with multiple goal modalities and proposed goal-aware action modeling pretraining and an actor-critic RL pipeline with distance-based extrinsic reward. Despite progress, these methods localize the goal guided by extrinsic reward, which might provide misleading guidance when the target’s location is unknown. Moreover, this approach fails to effectively and comprehensively model the environment, leading to inefficient search strategies and generalization to unseen regions. GeoExplorer aims to address these challenges by modeling action-state dynamics simultaneously and enabling curiosity-driven exploration.

Transformers for Action-State Modeling. Recent progress in RL has shown exciting potential in predicting actions through sequence modeling [7, 13] based on causal Transformers [23, 34]. They focus on predictive modeling of action sequences conditioned on a task specification (*e.g.*, target goal or returns) as opposed to explicitly learning Q-functions or policy gradients [9, 27, 29]. Moreover, recent work has focused on learning *world models* to describe environments. They further use transformer-style architecture to model the effect of the agent’s action on the environment, by predicting future state representations [3, 32, 40] or observations [36, 38]. They often take the previous state history and a potential action as input and predict the next state. However, those world models are primarily designed for domains with *continuous state transitions*, such as autonomous driving and robotics. In contrast, the AGL setting may involve non-overlapping and abrupt state changes by taking an action. GeoExplorer demonstrates that even in such cases, state modeling enhances action prediction, further underscoring the critical role of comprehensive environment modeling.

Curiosity-Driven Reinforcement Learning. Curious exploration plays a critical role in RL by enabling agents to navigate and learn in environments where rewards are far in the future, sparse, or non-existent. Even when the ex-

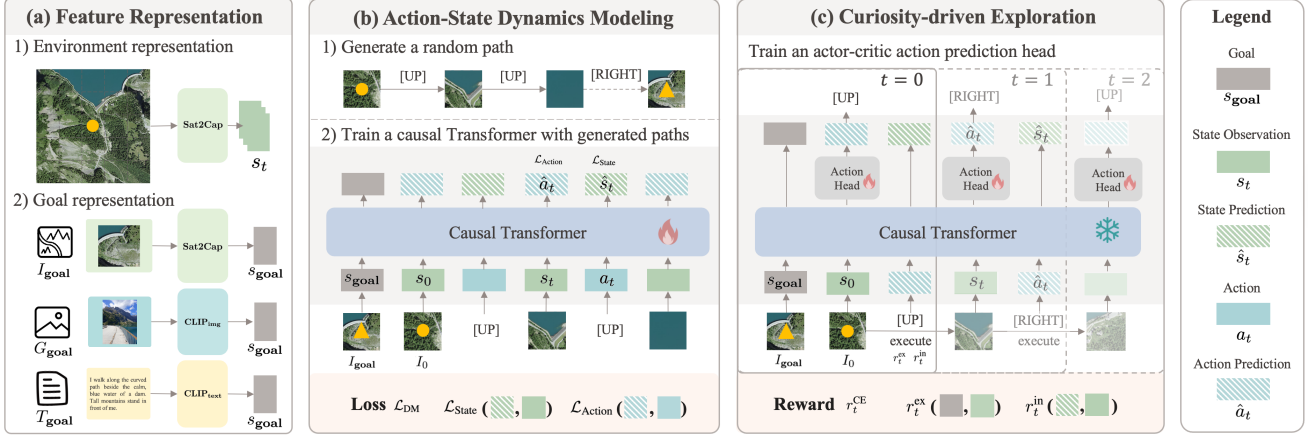


Figure 3. **GeoExplorer learning pipeline.** The learning process can be divided into three stages sequentially: Feature Representation, Action-State Dynamics Modeling (DM), and Curiosity-Driven Exploration (CE). (a) *Feature Representation.* The environment (s_t) and goal (s_{goal}) are encoded with different but aligned encoders, according to their modalities (e.g., aerial images (I_{goal}), ground-level images (G_{goal}), or text (T_{goal})). (b) *Action-State Dynamics Modeling.* A causal Transformer is trained to jointly capture action-state dynamics, guided by supervision from generated action-state trajectories for environment modeling. (c) *Curiosity-Driven Exploration.* Based on state prediction from (b), a curiosity-driven intrinsic reward (r_t^{in}) is combined with the distance-based extrinsic reward (r_t^{ex}) to encourage the agent to explore the environment by measuring the differences between prediction and observations.

intrinsic goal is well-defined, researchers have argued for the use of intrinsic motivation as an additional reward encouraging exploration [26]. Different approaches to generate intrinsic rewards have been explored, including encouraging exploration of previously unencountered states [6, 20], fostering curiosity when predictions about future states and the outcomes of actions differ [20, 30], promoting exploration in areas of high uncertainty [14] and encouraging agents to align their behaviors with the expectations of others [15]. Building upon these advances, we introduce curiosity-driven exploration to AGL, where adaptive and active exploration is essential for generalization. By rewarding the disagreement between predicted and actual states, GeoExplorer exemplifies the effective integration of curiosity-driven RL into sequence modeling frameworks, without additional components.

3. GeoExplorer

3.1. Overview

Problem Definition. Following previous work [21, 25], the AGL task can be considered as a goal-reaching RL problem. A UAV agent is located at a starting patch (I_0) and aims at navigating within a predefined non-overlapping $X \times Y$ search grid to reach a goal location within a search budget ($\mathcal{B}$). At each time step t , the agent observes only its current patch I_t and can move between patches by taking actions $a_t \in \mathcal{A}$ (UP, DOWN, LEFT, RIGHT, when considering a bird’s-eye-view). The state of time step t can be denoted as s_t , which is a representation of the current patch. The overall objective of AGL is to determine a trajectory

$x = \{s_0, a_0, \dots, s_t, a_t, \dots, s_{\text{goal}}\}$ that leads the agent to the goal. During training, the goal is exclusively presented as an aerial view (I_{goal}) with known location. For inference, the goal can be provided in one of the three modalities: aerial view, ground images (G_{goal}) or text (T_{goal}), but the location of the goal is unknown.

Method Overview. The training process of GeoExplorer can be divided into three stages (Figure 3): 1) *Feature Representation*, 2) *Action-State Dynamics Modeling (DM)*, and 3) *Curiosity-Driven Exploration (CE)*. DM is a supervised pretraining stage, where action-state dynamics are modeled by a causal Transformer. CE is a reinforcement learning stage, where an actor-critic network, used as an action prediction head, is trained to predict actions based on the output of the DM stage, combining both distance-based reward and curiosity-driven reward.

3.2. Feature Representation

In AGL, the goals are presented in different modalities. To achieve localization regardless of their nature, pre-aligned feature encoders are used for each modality (i.e., aerial images, ground-level images, and text), as in previous work [25] and as depicted in Figure 3 (a). Those modality-specific encoders are pre-trained to align within a joint embedding space, with text-image pairs or image-image pairs through contrastive learning [8, 24]. More specifically, the ground image encoder (a Vision Transformer, CLIP_{img}) and text encoder (CLIP_{text}) are issued from the pre-trained CLIP model [24], while the aerial encoder (Sat2Cap_{img}) is a ViT aligned with the CLIP image en-

coder using ground-level image and aerial image pairs [8]. The process can be denoted as:

$$\begin{aligned} s_{\text{goal}} &= \text{Sat2Cap}_{\text{img}}(I_{\text{goal}}), \\ \text{or } s_{\text{goal}} &= \text{CLIP}_{\text{img}}(G_{\text{goal}}), \\ \text{or } s_{\text{goal}} &= \text{CLIP}_{\text{text}}(T_{\text{goal}}). \end{aligned} \quad (1)$$

Moreover, we also encode the environment using the same aerial image encoder, and the agent’s location at time step t , patch I_t , can be encoded as current state s_t as: $s_t = \text{Sat2Cap}_{\text{img}}(I_t)$.

3.3. Action-State Dynamics Modeling

Different from previous efforts that use causal Transformers (CausalTrans, *e.g.*, LLMs) to predict *actions only* [7, 25], we simultaneously model the *action-state dynamics* without modifying the model architecture.

Random Path Generation. To collect action-state sequences for training, for each search area, we generate trajectories with random {start, goal} pairs, as in [25] (shown in Figure 3 (b)). Each generated sequence, denoted as $x_n = \{s_0, a_0, s_1, a_1, \dots, s_t, a_t, \dots, s_N, a_N\}$, consists of a series of randomly chosen actions $\{a_0, a_1, \dots, a_t, \dots, a_N\}$ that explore possible transitions between states. Additionally, we define a ground truth action transitions $\{a_0^*, a_1^*, \dots, a_t^*, \dots, a_N^*\}$ for each trajectory. Each action a_t^* represents an ideal goal-directed action which moves the agent from patch I_t closer to the goal patch I_{goal} .

Causal Transformer for Sequence Modeling. With the generated trajectories, we train a causal Transformer on a sequence modeling task that **1)** predicts the optimal actions a_t^* that will bring the agent closer to the goal location; **2)** predicts a state representation s_t at time t that depends on the action at the previous time step a_{t-1} . The modeling process, illustrated in Figure 3 (b), can be denoted as:

$$\hat{a}_t = \text{CausalTrans}(s_t | x_{t-1}, s_{\text{goal}}), \quad (2)$$

$$\hat{s}_t = \text{CausalTrans}(a_{t-1} | x_{t-1}, s_{\text{goal}}). \quad (3)$$

Before being processed by the causal Transformer, states are encoded using relative position encoding following [25].

Action Modeling Loss. We use the binary cross-entropy loss to guide the action prediction at each time step t along the action-state sequence, as in previous work [25]:

$$\begin{aligned} \mathcal{L}_{\text{Action}} &= \sum_{t=0}^N -(y_t \log(P_t) + (1 - y_t) \log(1 - P_t)), \\ P_t &= \text{Softmax}(\hat{a}_t), \end{aligned} \quad (4)$$

where N is the length of the random sequence. P_t is the predicted probability of actions at t when x_{t-1} and s_{goal} are given as the input to the model, and $y_t = [y_t^{(1)}, y_t^{(2)}, \dots, y_t^{(|\mathcal{A}|)}]$, indicates if an action from the $\mathcal{A}$ set

is considered optimal at time step t . $y_t^{(j)} = 1$ if the j ’th action leads agent closer to the goal, and $y_t^{(j)} = 0$ otherwise.

State Modeling Loss. Without modifying the model architecture, we extend the causal Transformer from predicting only actions to jointly predicting both states and actions. We do so by adding a state modeling loss. Inspired by recent advances in world models [3, 12, 40], we propose to use the mean squared error between the predicted state representation and the ground truth state representation to guide the model learning:

$$\mathcal{L}_{\text{State}} = \sum_{t=1}^{N-1} \|\hat{s}_t - s_t\|_2^2. \quad (5)$$

Note that here the total sequence length is $N - 1$ since we ignore the starting and ending state for prediction.

The final DM loss ($\mathcal{L}_{\text{DM}}$) is the combination of the action modeling loss ($\mathcal{L}_{\text{Action}}$) and the state modeling loss ($\mathcal{L}_{\text{State}}$):

$$\mathcal{L}_{\text{DM}} = \mathcal{L}_{\text{Action}} + \alpha \mathcal{L}_{\text{State}}, \quad (6)$$

where α is a scaling factor.

3.4. Curiosity-Driven Exploration

In this subsection, we first introduce the actor-critic pipeline for reinforcement learning, which serves as the foundation for our approach. Building on it, we then propose the curiosity-driven exploration guided by intrinsic rewards.

Actor-Critic Reinforcement Learning Pipeline. During CE, the pre-trained causal Transformer is frozen. In this stage, we introduce an action prediction head as shown in Figure 3 (c), which includes an *actor* $\pi_\theta(a|s_t)$ (policy network) and a *critic* $v_\psi(s_t)$ (value function) optimized by Proximal Policy Optimization (PPO) following the actor-critic reinforcement learning pipeline [28]. At each time step t , **(1)** the actor chooses an action $\hat{a}_t \sim \pi_\theta(a|s_t)$. **(2)** by executing $\hat{a}_t$, the state transits to s_{t+1} and the reward r_t^{CE} is obtained. **(3)** the value $v_\psi(s_t)$ denotes the expected total reward from state s_t is calculated by the critic. **(4)** the actor and critic are updated under PPO. See *supplementary materials* (S2) for details of the PPO agent. The agent operates on outputs of the causal Transformer and thus benefits from the action-state dynamics learned during DM.

Curiosity-Driven Exploration. The most straightforward reward for goal-reaching RL problems is designed based on whether the goal is achieved or not. However, these rewards are usually very sparse, leading to insufficient feedback to learn efficiently and effectively [20]. Previous efforts proposed shaping the reward to be dense, and feasible to obtain during the process. In particular, existing methods in AGL used an extrinsic reward based on the distance between the current and the goal patch, to optimize the actor-critic action prediction head, which can be denoted as:

$$r_t^{\text{ex}} = \begin{cases} 1, & \text{if } \|p_{t+1} - p_{\text{goal}}\|_2^2 < \|p_t - p_{\text{goal}}\|_2^2 \\ -1, & \text{if } \|p_{t+1} - p_{\text{goal}}\|_2^2 > \|p_t - p_{\text{goal}}\|_2^2 \\ 2, & \text{if } s_{t+1} = s_{\text{goal}} \end{cases} \vee s_{t+1} \in \{x_t\}, \quad (7)$$

where p_t denotes the position of patch I_t . The intuition behind the distance-based reward is *learning from minimizing the distance*: The extrinsic reward encourages the agent when its actions reduce the distance to the goal, with the highest reward given upon successfully reaching the target location. Conversely, the agent will be penalized for moving away from the goal or revisiting previously explored states. However, when the goal location is *unknown* (during inference), this reward becomes difficult to estimate and impacts the actor-critic network negatively.

To address this challenge, while encouraging exploration of the environment, we introduce intrinsic curiosity-driven rewards to AGL. Intrinsic rewards are designed following the intuition that an action is rewarded when it tends to explore an unexpected state. The degree of “unexpectedness” is measured based on the differences between state observation and prediction, the latter being built upon the state dynamics modeling in DM. Following the literature [15, 20], we define the curiosity-driven intrinsic reward for AGL based on two different measurements:

- **Mean squared error.** The reward is measured by the mean squared error between the predicted state representation ($\hat{s}_{t+1}$) and the observed state representation (s_{t+1}):

$$r_t^{\text{in}} = \|\hat{s}_{t+1} - s_{t+1}\|_2^2. \quad (8)$$

- **Cosine similarity.** The reward is measured by the cosine similarity between the two features.

$$r_t^{\text{in}} = -\cos(\hat{s}_{t+1}, s_{t+1}). \quad (9)$$

To combine the two sources of feedback (goal and the exploration), the final reward is a weighted sum of extrinsic and intrinsic rewards, weighted by a parameter β :

$$r_t^{\text{CE}} = r_t^{\text{ex}} + \beta r_t^{\text{in}}. \quad (10)$$

Note that before multiplying by β , we normalize intrinsic rewards to $[-1, 1]$.

3.5. Inference

During inference, given a goal presented as one of the three modalities and a starting patch, GeoExplorer represents the goal and environment, and determines a trajectory of actions and states $x = \{s_0, \hat{a}_0, \dots, s_t, \hat{a}_t, \dots, s_M\}$ of length M within a search budget $\mathcal{B}$ ($M \leq \mathcal{B}$) using the action prediction head. The actions are selected according to the argmax policy, *i.e.*, the highest probability, for validation. For path visualization, we use stochastic policy, *i.e.*, selecting action probabilistically, to show more possible paths.

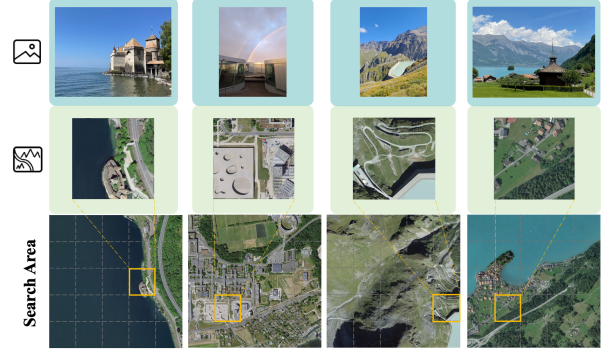


Figure 4. Examples from the proposed SwissViewMonuments dataset with unseen targets. From top to bottom: ground-level images, aerial goal patches, and search areas.

4. Data and Setup

Datasets. We consider four AGL benchmarks: **The Masa dataset** [18] uses aerial views as goals. The dataset contains 1188 images, with a split of 70% for training and 15% each for validation and test [21, 25]. Each image is regarded as a search area, and we randomly generate for each one 5 {start, goal} pairs to evaluate the model, resulting in 895 configurations for testing. **The MM-GAG dataset** [25] represents goals with different modalities such as text and ground-level images. It includes 73 search areas. **The xBD dataset** [11] is a subset of the xBD dataset test set. Following previous work [25], 800 images are kept for both before (xBD-pre) and after (xBD-disaster) disaster images. Note that to mimic real-world search-and-rescue missions, the goal is always represented with the pre-disaster image (xBD-pre), regardless of whether the xBD-pre or xBD-disaster is used as search areas. Our proposed **SwissView dataset** (Figure 4) is constructed from Swisstopo’s SWISSIMAGE 10cm imagery, with two distinct components: SwissView100, which comprises 100 images randomly selected from across the Swiss territory, thereby providing diverse natural and urban environment; and SwissViewMonuments, which includes 15 images of atypical or distinctive scenes, such as landmarks and unseen landscapes, with corresponding ground level images. Since the targets in the latter are unique, it is used to assess models’ generalization to unseen targets. Detailed descriptions can be found in the *supplementary materials* (S1). Following previous methods [21, 25], and unless specified otherwise, GeoExplorer is only trained on the Masa dataset and evaluated on the others.

Evaluation Metrics. We use the *success ratio* (SR) and *step-to-the-goal* (SG) for evaluation. Within a search budget $\mathcal{B}$, if the agent reaches the goal, the attempt is considered successful. SR is defined as the ratio of successful localizations to the total number of attempts. SG is the Manhattan distance between the path-end and goal locations. The eval-

uation is carried out over different distances C from the start to the goal location. The results of SG are provided in the supplementary materials (S4.7).

Evaluation Settings. We evaluate GeoExplorer in four settings: 1) *Validation*: we evaluate the model on the same dataset as it is trained. 2) *Cross-domain transfer*: the model trained on the Masa dataset is evaluated on an unseen dataset using aerial view as goal modality. 3) *Cross-modal generalization*: the goal is presented from different modalities. 4) *Unseen target generalization*: to evaluate the models’ adaptation to the unseen targets, we evaluate the model on our SwissView dataset, with aerial view and ground-level images as goals.

Implementation Details. Following [25], the search grid dimensions X and Y are set to 5, leading to 25 non-overlapping patches in a search area. The search budget B is set to 10 and the start-to-goal distance is chosen in a set $C \in \{4, 5, 6, 7, 8\}$. α (Eq. 6) is set to 1 and β (Eq. 10) is set to 0.25 according to parameter analysis (see in *supplementary materials S4.1*). Eq. 8 is the default setting of intrinsic reward. We use the pretrained CLIP vision (ViT-b-32) and text encoders [22] as encoders for ground-level images and text descriptions. For aerial images, we use the Sat2Cap satellite encoder [8], which is fine-tuned to align with the CLIP vision encoder using contrastive learning. The pre-trained Falcon-7B model [1] is used as the causal Transformer in GeoExplorer, similarly to [25]. The hyperparameter settings and model chosen follow the baselines to ensure a fair comparison. Detailed descriptions can be found in the *supplementary materials (S2)*.

Baselines. We compare GeoExplorer against:

- *Random policy*, which randomly selects an action at each time step; *PPO policy* [28] selects actions based on the current observation.
- *Decision Transformer (DiT)* [7], an RL pipeline trained with offline optimal trajectories with randomly generated {start, goal} pairs via action modeling.
- *AiRLoc* [21], an RL-based model designed for unimodal AGL tasks. A goal-related sparse reward is used to guide model learning.
- *GOMAA-Geo* [25], an RL-based method agnostic to multiple goal modalities for AGL tasks. It involves pretraining based on action modeling and an RL stage guided by distance-based extrinsic rewards.

5. Results

5.1. Numerical Results in the Evaluation Settings

Validation. We first evaluate the models on the Masa dataset, *i.e.*, the dataset used for training. Table 1 demonstrates that GeoExplorer outperforms all other comparison methods on the Masa dataset, particularly for longer search paths. Specifically, it achieves a success ratio improvement

Table 1. Numerical results on the test set of the **Masa dataset**. The success ratio (SR) is reported with different start-to-goal distance C . The results of the baselines are taken from [25].

Method	$C = 4$	$C = 5$	$C = 6$	$C = 7$	$C = 8$
Random policy	0.1412	0.0584	0.0640	0.0247	0.0236
PPO policy [28]	0.1427	0.1775	0.1921	0.2269	0.2595
AiRLoc [21]	0.1786	0.1561	0.2134	0.2415	0.2393
DiT [7]	0.2011	0.2956	0.3567	0.4216	0.4559
GOMAA-Geo [25]	0.4090	0.5056	0.7168	0.8034	0.7854
GeoExplorer	0.4324	0.5318	0.8156	0.9229	0.9497

Table 2. **Generalization across domain and goal modalities on the MM-GAG dataset**. The goal is presented as an aerial image (“I”), a ground-level image (“G”), or a text (“T”). Note that the models are only trained on the Masa dataset.

Goal	Method	$C = 4$	$C = 5$	$C = 6$	$C = 7$	$C = 8$
I	Random policy	0.1412	0.0584	0.0640	0.0247	0.0236
	PPO policy [28]	0.1489	0.1854	0.1879	0.2176	0.2432
	AiRLoc [21]	0.1745	0.1689	0.2019	0.2156	0.2290
	DiT [7]	0.2023	0.2856	0.3516	0.4190	0.4423
	GOMAA-Geo [25]	0.4085	0.5064	0.6638	0.7362	0.7021
	GeoExplorer	0.4338	0.5415	0.7631	0.8369	0.8277
G	GOMAA-Geo [25]	0.4383	0.5150	0.6808	0.7489	0.6893
	GeoExplorer	0.4308	0.5138	0.7200	0.8246	0.7815
T	GOMAA-Geo [25]	0.4000	0.4978	0.6766	0.7702	0.6595
	GeoExplorer	0.4431	0.4892	0.7200	0.8062	0.7631

of 0.0234 when $C = 4$ and 0.1643 when $C = 8$ compared to the baseline (GOMAA-Geo). Regarding reward design, approaches employing dense rewards, such as GOMAA-Geo or GeoExplorer demonstrate a significant advantage over the baseline utilizing sparse rewards (*e.g.*, AiRLoc). This highlights the importance of dense and timely guidance in the AGL process. Interestingly, we observe higher performance for larger C , even though we consider the task to be more challenging. This can be attributed to the limited variability in configurations in these scenarios, which makes it easier for the model to overfit to the training data distribution. (see *supplementary materials (S5.1)* for details).

Cross-Domain Transfer. For an AGL agent, cross-domain transferability is crucial, as models are typically pre-trained on one dataset and subsequently deployed in an *unseen environment*. Here, we provide the results of the models trained on the Masa dataset and then deployed on two other datasets, but again with aerial view as the goal modality. For MM-GAG (Table 2), when the goal is presented as an aerial view (the first block ‘Goal = I’), GeoExplorer leads to the best overall performance, with a notable improvement of 0.0772 on average. Similarly, the results for the xBD dataset (shown in Table 3) confirm the effectiveness of GeoExplorer in cross-domain transfer setting, with an average improvement of 0.0556 on the xBD-disaster subsets. Note that this setting is extremely challenging since the goal is presented as pre-disaster images.

Cross-Modal Generalization. In real-world applications,

Table 3. **Cross-domain transfer** on the **xBD-disaster dataset**. Note that the models are only trained on the Masa dataset and the goal is always presented from the aerial view before the disaster.

Method	$C = 4$	$C = 5$	$C = 6$	$C = 7$	$C = 8$
Random policy	0.1412	0.0584	0.0640	0.0247	0.0236
PPO policy [28]	0.1132	0.1146	0.1292	0.1665	0.1953
AiRLoc [21]	0.1201	0.1298	0.1507	0.1631	0.1989
DiT [7]	0.1012	0.2389	0.3067	0.3390	0.3543
GOMAA-Geo [25]	0.4002	0.4632	0.6553	0.7391	0.6942
GeoExplorer	0.3975	0.5025	0.7185	0.8190	0.7923

Table 4. **Unseen objects generalization ability** on the **SwissViewMonuments dataset**. * corresponds to results obtained using the pretrained model [25].

Method	$C = 4$	$C = 5$	$C = 6$	$C = 7$	$C = 8$
I GOMAA-Geo*	0.4027	0.3833	0.6267	0.7278	0.7833
I GeoExplorer	0.4133	0.5333	0.7700	0.8889	0.8833
G GOMAA-Geo*	0.4107	0.3833	0.6367	0.7722	0.7500
G GeoExplorer	0.4160	0.5167	0.7733	0.8778	0.7833

Table 5. **Ablation study** on the Masa dataset on state modeling and reward design. “MSE” and “COS” denote intrinsic reward based on mean squared error or cosine similarity, respectively.

$\mathcal{L}_{\text{Action}}$	$\mathcal{L}_{\text{State}}$	r_t^{ex}	r_t^{in}	$C = 4$	$C = 5$	$C = 6$	$C = 7$	$C = 8$
✓		✓		0.4090	0.5056	0.7168	0.8034	0.7854
✓	✓	✓		0.3978	0.4939	0.7609	0.8413	0.8648
✓		✓	MSE	0.3888	0.4938	0.7408	0.8615	0.9140
✓	✓	✓	COS	0.3955	0.5385	0.8134	0.9050	0.9352
✓	✓	✓	MSE	0.4324	0.5318	0.8156	0.9229	0.9497

the goal may be described using other modalities than aerial images. Experimental results on the MM-GAG dataset presented in Table 2 suggest that with the same modality-specific encoders as used in GOMAA-Geo, GeoExplorer still shows better generalizability when the goal is described in ground-level images (second block ‘Goal = G’) and text (the third block ‘Goal = T’). With less direct information, GeoExplorer achieves performances comparable to the baseline on short paths, while significantly improving SR when the path is longer (0.0690 (ground-level images) and 0.0610 (text) on average for $C = \{6, 7, 8\}$).

Unseen Target Generalization. To assess the generalization ability of GeoExplorer and the baseline models, we design an *unseen target* generalization experiment on the SwissViewMonuments dataset. Unlike previous datasets, where goal patches were randomly selected, this dataset has fixed goal locations. As suggested in Table 4, GeoExplorer exhibits an impressive generalization ability in localizing unseen targets, especially when the path is long (e.g., for $C = 6$, improving 0.1433 when using aerial view and 0.1366 when using ground-level images). More results from the SwissView dataset and evaluation using SG as metric are presented in *supplementary materials (S4)*.

5.2. Ablation Study

We conduct ablation studies on the Masa dataset: first, we vary the training objectives in the DM stage to analyze the impact of the proposed Action-State Dynamics Modeling strategy (Eq. 6). Comparing Row 2 with Row 1, by adding state modeling loss, SR improves 0.0277 on average, which demonstrates the effectiveness of state transition modeling on action prediction, especially for longer distances. Adding intrinsic reward without state modeling also shows improvements (0.0357 on average, Row 3 and 1), which indicates that state prediction without explicit supervision, even though inaccurate, still provides useful information for exploration. GeoExplorer with action-state dynamics modeling yields the best overall performance. With a learned state modeling, SR increases 0.0507 in average (Row 5 and 3). We also test different configurations of the reward design. Results show similar performance for the two reward designs, with an average SR difference of 0.0130. We noticed a slight performance drop on short paths (Row 1 and 2/3). For shorter paths, the agent has limited steps to the goal, leaving less room for exploration, as it may lead to detours. Additionally, state modeling on short paths may be inaccurate due to limited history state observations.

5.3. Does Intrinsic Reward Improve Exploration?

Visualization of Exploration Ability. To better understand the exploration capabilities of GeoExplorer and the baseline GOMAA-Geo, we visualize the generated paths in Figure 5. In panel (a), we compare {start, goal} pairs with similar semantic content from two different search areas. When the distance between the start and the goal is short and easy to estimate, GOMAA-Geo successfully reaches the goal with the shortest path (upper example). However, in the lower example, where visually similar start and goal patches are located in different parts of the image, GOMAA-Geo struggles to accurately localize the goal, as it finds it more challenging to estimate the potential distance. In contrast, GeoExplorer succeeds, leveraging its action-state modeling ability to differentiate between possible paths. In Figure 5 (b), we reverse the {start, goal} patches to evaluate the model’s robustness. GOMAA-Geo fails to maintain robustness, while GeoExplorer benefits from its exploration ability and successfully reaches the target irrespective of the search direction. The visited patch distribution shown in Figure 5 (c) provides a more comprehensive observation of the exploration ability: when $C = 4$, only 20.08% of the visited patches are located in the central part of the search area for GOMAA-Geo, while GeoExplorer increases this ratio to 30.79%, indicating a more comprehensive exploration. Those findings further confirm that Geoexplorer shows robust, diverse, and content-related exploration ability.

Analysis of Curiosity Patterns. We provide an in-depth analysis of the curiosity-driven intrinsic reward. In Figure 6

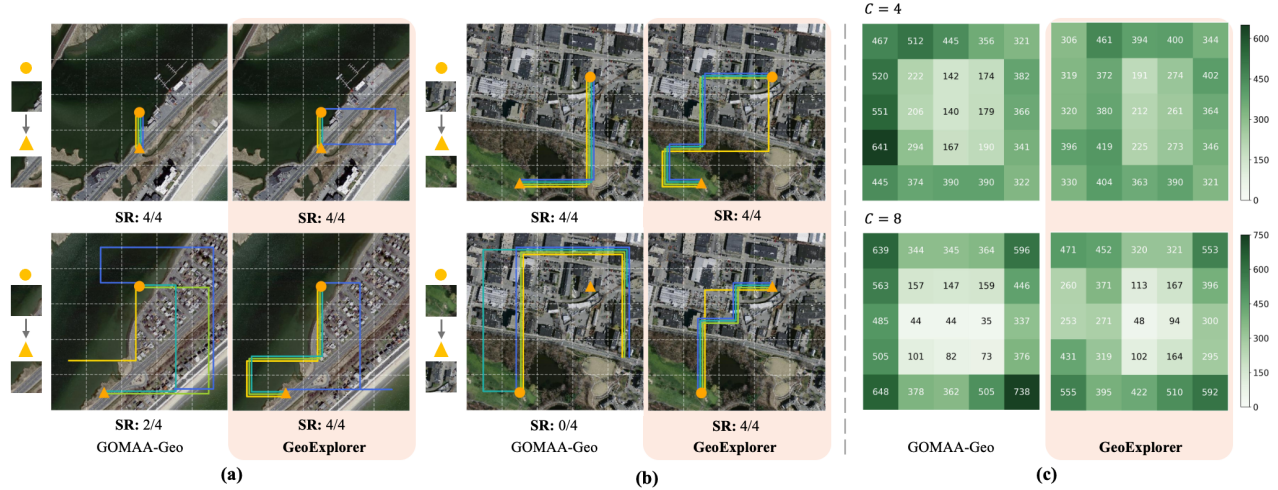


Figure 5. **GeoExplorer shows robust, diverse, and content-related exploration ability.** Given a pair of {start (o), goal (Δ)} per search area, models generate four trials with stochastic policy (Sec. 3.5), randomly shown in four different colors. (a) Similar {start, goal} pairs with different distance. Despite the difficulties of estimating distance, GeoExplorer’s exploration ability maintains robust and diverse. (b) Reversed {start, goal} pairs in the same search area. GeoExplorer’s exploration is adaptive and robust to goal content irrespective of the search direction. (c) Statistics of visited patches by all generated paths for both models when the distance $C = 4$ and $C = 8$ in Masa test set. Numbers denote the counts of visits per patch. GeoExplorer develops a more comprehensive exploration covering the search area.



Figure 6. **Curiosity-driven intrinsic rewards provide dense, goal-agnostic, and content-related guidance to enhance the exploration ability of GeoExplorer.** Examples from the MM-GAG dataset targeting multiple goal modalities. (a) The goal of each example (column) is represented in different modalities. (b) Four trials, as well as average intrinsic rewards along the paths. (c) Average intrinsic reward from four trials at the patch level.

(b), we show four paths retrieved by our model and their corresponding average accumulated intrinsic reward (in the same colors as the path). As expected, we observe that trajectories visiting diverse image content (e.g., forest, city and river as in the green and blue trajectories) present higher

average intrinsic rewards than those crossing forest patches only (e.g., the yellow trajectory). Figure 6 (c) presents the curiosity scores per patch. Note that for visualization, we normalize the intrinsic rewards. Curiosity-reward is naturally dense and content-related. Patches with higher intrinsic reward turn out to be more “interesting”, i.e., the semantic content of these patches can hardly be predicted from the surrounding patches. For instance, the patch with the highest score in the right column example depicts the transition from a forest to an urban environment. These results show that curiosity-driven rewards provide dense, goal-agnostic, and content-related guidance, enhancing the exploration ability of GeoExplorer.

6. Conclusion

Adaptive and active exploration is a step toward developing AGL in real-world search-and-recue operations. In this paper, we propose GeoExplorer, an AGL agent that explores a search region with a combination of goal-oriented and *curiosity-driven* guidance based on comprehensive *environment modeling*. The curiosity-driven reward is goal-agnostic and provides dense and reliable feedback by leveraging the discrepancy between the state prediction and observation to encourage exploration. Extensive experiments and analyses on four AGL benchmarks demonstrate that GeoExplorer outperforms state-of-the-art methods and exhibits remarkable transferability to unseen targets and environments, highlighting its robust, generalizable, content-aware exploration ability driven by intrinsic curiosity.

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